Numerical Methods in Python: Structure & Reasoning

Example Problem: Radioactive Decay

ODE: $\frac{dN}{dt} = -kN$

Initial Condition: $N(0) = N_o$

We'll solve this numerically using Euler's method

Step 1: Import Required Packages

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

Step 2: Define Constants & Initial Conditions

$N_0 = 1000$ # Initial number of nuclei

k = 0.1 # Decay constant

dt = 0.1 # Time step (s)

Step 3: Initialize list/ Arrays

$N = [N_0]$

$t = [0]$

Step 4: Choose Loop Type

Use while-loop: run until N is small (e.g., <1)

Use for-loop: fixed simulation time (e.g., 100 steps)

Step 5: Euler Update Rule

```
dNdt = -k * N[-1]
```

```
N_next = N[-1] + dNdt * dt
```

Step 6: Append to lists/ Arrays

```
N.append(N_next)
```

```
t.append(t[-1] + dt)
```

Step 7: Plotting Results

```
plt.plot(t, N)
```

```
plt.xlabel('Time (s)')
```

```
plt.ylabel('Nuclei Remaining')
```

```
plt.title('Radioactive Decay using Euler Method')
```

Summary: Programming Structure

